

Problem Set 1

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1.

a)

Treatment	Favorable	Unfavorable	Total
Placebo	34	61	95
Test	51	39	90
Total	85	100	185

Table 1: 2 x 2 Contingency Table for Treatment and Response

b)

The FREQ Procedure

Table of treat by outcome

treat outcome

Frequency			
Row Pct	F	U	Total
-----+-----+-----+			
test	51	39	90
	56.67	43.33	
-----+-----+-----+			
placebo	34	61	95
	35.79	64.21	
-----+-----+-----+			
Total	85	100	185

Statistics for Table of treat by outcome

Column 1 Risk Estimates

	Risk	ASE	95% Confidence Limits		Exact 95% Confidence Limits	
Row 1	0.5667	0.0522	0.4587	0.6746	0.4580	0.6708
Row 2	0.3579	0.0492	0.2562	0.4596	0.2621	0.4628
Total	0.4595	0.0366	0.3849	0.5340	0.3861	0.5341
Difference	0.2088	0.0717	0.0573	0.3602		

Difference is (Row 1 - Row 2)

The asymptotic confidence limits include a continuity correction.

Confidence Limits for the Risk Difference

Risk Difference = 0.2088

Type	95% Confidence Limits	
Newcombe (Corrected)	0.0574	0.3480

Column 1 (outcome = F)

The proportion of participants with favorable response in the placebo group is 0.3759 (0.2621, 0.4628). The proportion of participants with favorable response in the test group is 0.5667 (0.4580, 0.6708).

c) The difference of proportions estimate for Test vs. Placebo for the Favorable response is 0.2088 with confidence limits (0.0574, 0.3480).

d) All expected counts have to be greater than 5, ideally greater than 10. The data should be count data and variables should be mutually exclusive.

Chi-square test will be used. The null hypothesis is that there is no association between treatment group and response. Pearson's chi-square is 8.11 ($p = 0.0044$). As this p-value is < 0.05 , we would reject the null hypothesis. The output in 1b) shows that there is a strong association between treatment and outcome such that test treatment produces a more favorable response than placebo treatment. Test treatment resulted in 57% favorable outcome while placebo treatment resulted in 36% favorable outcome.

Statistics for Table of treat by outcome

Statistic	DF	Value	Prob
Chi-Square	1	8.1108	0.0044
Likelihood Ratio Chi-Square	1	8.1677	0.0043
Continuity Adj. Chi-Square	1	7.2920	0.0069
Mantel-Haenszel Chi-Square	1	8.0669	0.0045
Phi Coefficient		-0.2094	
Contingency Coefficient		0.2049	
Cramer's V		-0.2094	

Treatment	Favorable	Unfavorable	Total
Placebo	85*95/185 = 43.65	95*100/185 = 51.35	95
Test	85*90/185 = 41.35	90*100/185 = 48.65	90
Total	85	100	185

e) The proportion of participants with favorable response in the placebo group is 0.3759 with exact confidence limits (0.2621, 0.4628). The proportion of participants with favorable response in the test group is 0.5667 with exact confidence limits (0.4580, 0.6708). The difference in proportion of participants with favorable response in the test group as compared to placebo group is 0.2088 with Newcombe corrected limits (0.0574, 0.3480). As 0 is not contained in this interval, we conclude that the difference in proportion of participants with favorable response between test and placebo group is statistically significant. Pearson's Chi-square had a value of 8.11 and $p = 0.0044$. With this p-value, we reject the null hypothesis and conclude that there is a statistically significant association between treatment group and response.

Answering Q2

*****;

*Entering in data;

data odds;

length center satisfactory \$1 treatment \$10;

infile cards dsd dlm='|' trunccover;

input center treatment satisfactory count;

cards;

A|High Dose|S|58

A|High Dose|U|17

A|Low Dose|S|25

A|Low Dose|U|30

B|High Dose|S|6

B|High Dose|U|2

B|Low Dose|S|3

B|Low Dose|U|7

;

proc print;

run;

options nocenter;

ods listing file="/home/lrackley0/BIOS 665/hw1.lst";

ods select crosstabfreqs RelativeRisks;

proc freq data = odds order = data;

where center='A';

tables treatment*satisfactory / chisq measures nocol nopct;

weight count;

run;

ods listing close;

ods listing file="/home/lrackley0/BIOS 665/hw1.lst";

ods select crosstabfreqs OddsRatioExactCL;

proc freq data = odds order = data;

where center = "B";

tables treatment*satisfactory / nocol;

weight count;

exact or;

run;

ods listing close;

options nodate nocenter;

ods listing file="/home/lrackley0/BIOS 665/hw1.lst";

proc freq data = odds order = data;

weight count;

where center='B';

tables treatment*satisfactory / chisq nocol expected;

exact chisq;

run;

ods listing close;

Table of treatment by satisfactory

treatment satisfactory

Frequency			
Row Pct	S	U	Total
-----+-----+-----+			
High Dose	58	17	75
	77.33	22.67	
-----+-----+-----+			
Low Dose	25	30	55
	45.45	54.55	
-----+-----+-----+			
Total	83	47	130

Odds Ratio and Relative Risks

Statistic	Value	95% Confidence Limits	

Odds Ratio	4.0941	1.9193	8.7331
Relative Risk (Column 1)	1.7013	1.2424	2.3298
Relative Risk (Column 2)	0.4156	0.2565	0.6734

a) The estimate of the odds ratio is 4.09. This means that the odds of satisfactory response to chronic eye irritation are 4.09 times more likely in the high dose group than low dose group.

b) The approximate 95% confidence limits for the odds ratio are (1.919, 8.733).

c) As 1 is not contained in the confidence interval, this indicates there is a significant association between treatment and response to chronic eye irritation.

d)

Table of treatment by satisfactory

treatment satisfactory

Frequency			
Percent			
Row Pct	S	U	Total
-----+-----+-----+			
High Dose	6	2	8
	33.33	11.11	44.44
	75.00	25.00	
-----+-----+-----+			
Low Dose	3	7	10
	16.67	38.89	55.56
	30.00	70.00	
-----+-----+-----+			
Total	9	9	18
	50.00	50.00	100.00

Odds Ratio

Odds Ratio	7.0000

Asymptotic Conf Limits

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95% Lower Conf Limit      0.8612
95% Upper Conf Limit      56.8946

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Exact Conf Limits
95% Lower Conf Limit      0.6219
95% Upper Conf Limit      99.8735

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Since counts are less than 10, we use exact OR. The odds ratio is 7.00 (0.6219,99.8735). High dose participants have 7 times higher odds of achieving satisfactory response to chronic eye irritation than low dose participants.

e) Fisher's exact test will be used since only 50% of the cells have expected counts > 5 .

H_0 : There is no association between treatment and response.

H_a : There is an association between treatment and response.

The FREQ Procedure

Table of treatment by satisfactory

treatment	satisfactory		
Frequency			
Expected			
Percent			
Row Pct	S	U	Total
-----+-----+-----+			
High Dose	6	2	8
	4	4	
	33.33	11.11	44.44
	75.00	25.00	
-----+-----+-----+			
Low Dose	3	7	10
	5	5	
	16.67	38.89	55.56
	30.00	70.00	
-----+-----+-----+			
Total	9	9	18
	50.00	50.00	100.00

Statistics for Table of treatment by satisfactory

Statistic	DF	Value	Prob

Chi-Square	1	3.6000	0.0578
Likelihood Ratio Chi-Square	1	3.7387	0.0532
Continuity Adj. Chi-Square	1	2.0250	0.1547
Mantel-Haenszel Chi-Square	1	3.4000	0.0652
Phi Coefficient		0.4472	
Contingency Coefficient		0.4082	
Cramer's V		0.4472	

WARNING: 50% of the cells have expected counts less than 5.

(Asymptotic) Chi-Square may not be a valid test.

Pearson Chi-Square Test

Chi-Square	3.6000
DF	1
Asymptotic Pr > ChiSq	0.0578
Exact Pr >= ChiSq	0.1534

Fisher's Exact Test

Cell (1,1) Frequency (F)	6
Left-sided Pr <= F	0.9924
Right-sided Pr >= F	0.0767
Table Probability (P)	0.0691
Two-sided Pr <= P	0.1534

Sample Size = 18

Since the two-sided p-value for Fisher's exact test is 0.1534, which is > 0.05 , we fail to reject the null and conclude that there is no significant association between treatment and response.

3.

a) McNemar's test has to be used because the data comes in matched pairs, namely an eye drop distributed to a left and right eye on the same patient. We compare a dichotomous condition of eye being clear or not clear of proteins for a single random sample from a single population.

$$Q_M = \frac{(22 - 55)^2}{22 + 55} = 14.1429$$

Table of placebo_resp by new_treat_resp

placebo_resp	new_treat_resp		
Frequency			
Row Pct	Clear	Not Clear	Total
-----+-----+-----+			
Clear	129	22	151
	85.43	14.57	
-----+-----+-----+			
Not Clear	55	44	99
	55.56	44.44	
-----+-----+-----+			
Total	184	66	250

McNemar's Test

Chi-Square	DF	Pr > ChiSq	Exact Pr >= ChiSq
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14.1429 1 0.0002 0.0002

$H_0: \pi_{12} = \pi_{21}$

$H_A: \pi_{12} \neq \pi_{21}$

When comparing this to χ^2_1 , this statistic is clearly significant ($p=0.0002$). We conclude that treatment had an effect on patients who cleared proteins in one eye and not the other.

b) This result is favorable for the new treatment because $\hat{p}_{new} - \hat{p}_{plac} = 0.132$. The proportion of patients clearing proteins in the eye under new treatment is greater than the proportion of patients clearing protein in the eye under placebo treatment.

$\hat{p}_{plac} = 151/250 = 0.604$ and $\hat{p}_{new} = 184/250 = 0.736$.

c) Here $\hat{p}_{plac} = 151/250 = 0.604$ and $\hat{p}_{new} = 184/250 = 0.736$. The estimated difference is $0.736 - 0.604 = 0.132$. The standard error is estimated to be

$$SE(\hat{p}_{new} - \hat{p}_{plac}) = \frac{1}{250} \sqrt{(22 + 55) - \frac{(22 - 55)^2}{250}} = 0.03409258$$

Thus, the confidence interval under the alternative is $0.132 \pm 1.96(0.03409258) = (0.06517854, 0.198821456)$.

4.

Disease	Test+	Test-	Total
Present	92	30	112
Absent	20	75	95
Total	112	105	217

Statistics for Table of disease by test

Column 1 Risk Estimates

	Risk	ASE	95% Confidence Limits		Exact 95% Confidence Limits	
Row 1	0.7541	0.0390	0.6777	0.8305	0.6679	0.8275
Row 2	0.2105	0.0418	0.1285	0.2925	0.1336	0.3062
Total	0.5161	0.0339	0.4496	0.5826	0.4475	0.5843
Difference	0.5436	0.0572	0.4315	0.6556		

Difference is (Row 1 - Row 2)

Column 2 Risk Estimates

	Risk	ASE	95% Confidence Limits		Exact 95% Confidence Limits	
Row 1	0.2459	0.0390	0.1695	0.3223	0.1725	0.3321
Row 2	0.7895	0.0418	0.7075	0.8715	0.6938	0.8664
Total	0.4839	0.0339	0.4174	0.5504	0.4157	0.5525

Difference -0.5436 0.0572 -0.6556 -0.4315

Difference is (Row 1 - Row 2)

- a) The sensitivity is 75.51%. Se = 92/122. The specificity is 78.95%. Sp = 75/95.
b) The exact 90% confidence limits for sensitivity are (0.6816,0.8171).

Statistics for Table of disease by test

Column 1 Risk Estimates

	Risk	ASE	90% Confidence Limits		Exact 90% Confidence Limits	
Row 1	0.7541	0.0390	0.6900	0.8182	0.6816	0.8171
Row 2	0.2105	0.0418	0.1417	0.2793	0.1441	0.2910
Total	0.5161	0.0339	0.4603	0.5719	0.4581	0.5738
Difference	0.5436	0.0572	0.4495	0.6376		

Difference is (Row 1 - Row 2)

- c) The exact 95% confidence limits for specificity are (0.1441,0.2910).
d) We will use Bayes' Rule to get NPV.

$$P(D^-|T^-) = \frac{[P(D^-) \times P(T^-|D^-)]}{[P(D^-) \times P(T^-|D^-)] + [P(D^+) \times P(T^-|D^+)]}$$

$$P(D^-|T^-) = \frac{0.30 \times 0.7895}{0.30 \times 0.7895 + 0.70 \times (1 - 0.7541)} = 57.91\%$$

5.

5a

The POWER Procedure

Fisher's Exact Conditional Test for Two Proportions

Fixed Scenario Elements

Distribution	Exact conditional
Method	Walters normal approximation
Group 1 Proportion	0.66
Group 2 Proportion	0.42
Nominal Power	0.9
Number of Sides	2
Alpha	0.05

Computed N per Group

Actual N per

Power Group

0.903 98

5b

The POWER Procedure

Fisher's Exact Conditional Test for Two Proportions

Fixed Scenario Elements

Distribution	Exact conditional
Method	Walters normal approximation
Group 1 Proportion	0.66
Group 2 Proportion	0.42
Group 1 Weight	2
Group 2 Weight	1
Nominal Power	0.8
Number of Sides	2
Alpha	0.05

Computed N Total

Actual	N
Power	Total
0.801	168

5c

The POWER Procedure

Fisher's Exact Conditional Test for Two Proportions

Fixed Scenario Elements

Distribution	Exact conditional
Method	Walters normal approximation
Group 1 Proportion	0.66
Group 2 Proportion	0.42
Group 1 Sample Size	70
Group 2 Sample Size	70
Number of Sides	2
Alpha	0.05

Computed Power

Power

0.771

a) We need 196 patients to achieve 90% power.

- b) The overall $N = 168$ so $n = 56$ in the placebo group and $n = 112$ in the test treatment group.
- c) We would achieve 77.1% power with 70 patients enrolled in each treatment group.